PROPERTY $(TT)/T$ AND COORDINATE-BLOCK FACTORIZATION OVER STRONGLY PURELY INFINITE RINGS

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ABSTRACT. We prove a coordinate-block factorization theorem for general linear groups over rings satisfying the single-sandwich condition

$$a \neq 0 \implies xay = 1 \text{ for some } x, y.$$

Every element of $\mathrm{GL}_n(R)$ is a product of at most $2n+6$ factors, each an elementary transvection or an element of one fixed coordinate copy of $\mathrm{GL}_{n-1}(R)$. This is not bounded elementary generation: the coordinate block is an essential part of the statement.

We combine this algebraic factorization with relative quasi-cocycle rigidity for elementary groups over finite free associative $\mathbb{F}_2$-algebras. If R is a nontrivial finite-type $\mathbb{F}_2$-algebra carrying a binary Leavitt family, satisfying the single-sandwich condition, and satisfying $\mathrm{diag}(u, 1) \in \mathrm{E}_2(R)$ for every $u \in R^\times$, then $\mathrm{E}_n(R)$ has property $(TT)/T$ for every $n \geq 3$. Consequently,

$$\mathrm{E}_n\big(L_{\mathbb{F}_2}(1,2)\big) \quad \text{has property } (TT)/T \qquad (n \geq 3).$$

We also give direct quantitative proofs of the relative root estimate and of the unstable diagonal reduction used in the specialization. The stable computation $K_1(L_{\mathbb{F}_2}(1,2)) = 0$, the GE property of purely infinite simple rings, and the general relative-rigidity strategy are due to earlier work; we isolate our contribution explicitly. Every stated result is matched to a theorem checked by the Lean kernel.

CONTENTS

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1. Introduction

Property $(TT)/T$, introduced by Mimura, asks that Hilbert-space quasi-cocycles be bounded after the invariant-vector part of the representation has been removed. It is stronger than Kazhdan's property (T) and is designed to control bounded cohomology and homomorphisms into groups admitting unbounded quasi-cocycles [5]. Ershov and Jaikin-Zapirain proved property (T) for elementary groups over arbitrary finitely generated associative rings [3]. The corresponding quasi-cocycle problem is subtler: the bounded error in the cocycle identity accumulates under a long word, so finite generation alone gives no global bound.

The present paper supplies a globalization mechanism for a class of strongly purely infinite rings. Its algebraic input is an explicit matrix factorization. The hypothesis is the following strong two-sided division property:

$$a \neq 0 \quad \Longrightarrow \quad \exists x, y \in R, \qquad xay = 1. \tag{1}$$

It is often called the single-sandwich property. It is available in the purely infinite simple setting relevant here [2].

Theorem A (Coordinate-block factorization)**.** *Let R be a nontrivial ring satisfying (1), let $n \geq 2$, and fix a coordinate $j < n$. Every $A \in \mathrm{GL}_n(R)$ admits a factorization of length at most $2n+6$ in which every factor is either an elementary transvection or belongs to the coordinate copy*

$$\mathrm{GL}_{n-1}^{(j)}(R) = \{B \in \mathrm{GL}_n(R) : Be_j = e_j,\ e_j^t B = e_j^t\}.$$

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coordinateBlock_factorization

Theorem A is deliberately not called bounded elementary generation. It is a relative bounded-width statement, with respect to elementary root subgroups and one fixed coordinate block. This is exactly the form required by the quasi-cocycle argument.

For the synthesis theorem, a *binary Leavitt family* in R means elements x_0, x_1, y_0, y_1 satisfying

$$y_i x_j = \delta_{ij}, \qquad x_0 y_0 + x_1 y_1 = 1.$$

It supplies the matrix self-similarity used to move between ranks.

Theorem B (Rigidity synthesis)**.** *Let R be a nontrivial finite-type $\mathbb{F}_2$-algebra. Assume that*

(1) R contains a binary Leavitt family;
(2) R satisfies the single-sandwich property (1);
(3) $\mathrm{diag}(u, 1) \in \mathrm{E}_2(R)$ for every $u \in R^\times$.

Then $\mathrm{E}_n(R)$ has property $(TT)/T$ for every $n \geq 3$.

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Here finite type is the fourth structural hypothesis: it is stated in the opening sentence rather than repeated in the numbered list. The Lean theorem has precisely these hypotheses. In particular, there is no residual-finiteness, soficity, approximation, or unstated regularity assumption in Theorem B.

Theorem C (Binary Leavitt algebra)**.** *For every $n \geq 3$,*

$$\mathrm{E}_n(L_{\mathbb{F}_2}(1, 2)) \quad \textit{has property } (TT)/T.$$

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Combining Theorem C with the independently known nonsoficity theorem gives the following striking, but logically secondary, consequence.

Corollary 1.1. *For every $n \geq 3$, the group $\mathrm{E}_n(L_{\mathbb{F}_2}(1, 2))$ is nonsofic and has property $(TT)/T$.*

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The contribution of this paper to Corollary 1.1 is the $(TT)/T$ assertion. The nonsoficity assertion is credited to the prior nonsofic-group construction and is used only as an external mathematical input to the corollary.

Credit and novelty boundary. Several ingredients have strong antecedents, and separating them is mathematically important.

- Mimura introduced property $(TT)/T$, developed its globalization strategy, and proved the relevant relative quasi-cocycle results for universal-lattice settings [5, 6].
- Ershov and Jaikin-Zapirain proved ordinary property (T) for elementary groups over finitely generated associative rings [3].
- Ara, Goodearl, and Pardo established the GE and unit-group K_1 structure for purely infinite simple rings [2].
- Ara, Brustenga, and Cortiñas computed the stable algebraic K-theory of Leavitt path algebras [1]. In particular, $K_1(L_k(1, 2)) = 0$ for a field k.

Accordingly, we do not claim the stable K_1-vanishing, the GE property, the bare equality $\mathrm{GL}_2(L_{\mathbb{F}_2}(1, 2)) = \mathrm{E}_2(L_{\mathbb{F}_2}(1, 2))$, or the abstract relative-rigidity architecture as new theorems. The main contributions isolated here are the explicit width $2n + 6$ coordinate-block factorization and its synthesis with quasi-cocycle control into Theorem B. The direct root and rank-two arguments give quantitative, self-contained routes to inputs which also have prior structural explanations.

Formal verification. The accompanying Lean development is available at https://github.com/SauersML/nonsofic_existence. Margin notes link theorem statements to declarations. The paper-facing declarations are collected in `PropertyTT/PaperStatements.lean`, `KOne/PaperStatements.lean`, and `PropertyTT/NonsoficCorollary.lean`. The endpoint audit reports only Lean's standard logical axioms propext, Classical.choice, and Quot.sound.

2. Quasi-cocycles and relative control

Let G be a group and let $\pi : G \to \mathcal{U}(\mathcal{H})$ be a unitary representation on a complex Hilbert space. A map $b : G \to \mathcal{H}$ is a *quasi-cocycle of defect at most* D if

$$\|b(gh) - b(g) - \pi(g)b(h)\| \le D \qquad (g, h \in G). \tag{2}$$

An exact cocycle has $D = 0$; it is the translation part of an affine isometric action. A quasi-cocycle permits uniformly bounded failure of the affine action law.

Definition 2.1. A group G has property $(TT)/T$ if, for every unitary representation π with no nonzero invariant vector, every Hilbert-space quasi-cocycle into π is bounded on G.

The local input used below is slightly stronger than a relative $(TT)/T$ statement because no hypothesis is imposed on invariant vectors.

Theorem 2.2 (Root quasi-cocycle estimate)**.** *Let X be finite, put $A = \mathbb{F}_2\langle X\rangle$, and let*

$$G = \mathrm{E}_4(A), \qquad X_{03} = \{I + aE_{03} : a \in A\}.$$

For every unitary representation π, every $D \ge 0$, and every quasi-cocycle $b : G \to \mathcal{H}$ of defect at most D, the set $b(X_{03})$ is bounded.

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`hasRelativeTT`

The theorem is stated in its stronger, invariant-vector-free form. Our proof is quantitative and formalized independently. We do not make a priority claim for the bare relative assertion: Mimura's noncommutative universal-lattice arguments are directly adjacent, and the characteristic-two fixed-space step below is short once the moving-space estimate is known.

2.1. **Fixed and moving components.** Let N be the last-column subgroup generated by X_{03}, X_{13}, X_{23}. Decompose

$$\mathcal{H} = \mathcal{H}^N \oplus (\mathcal{H}^N)^\perp, \qquad b(r) = p + w$$

for $r = I + aE_{03}$. Since the coefficient ring has characteristic two, $r^2 = 1$. Applying (2) to r^2, and projecting to $\mathcal{H}^N$, bounds the fixed component:

$$\|p\| \le D. \tag{3}$$

The moving component is controlled by two embedded rank-three planes. If s commutes with r, comparison of the quasi-cocycle equations for sr and rs gives

$$\|\pi(s)b(r) - b(r)\| \le 2\,\|b(s)\| + 2D.$$

The finite rank-three control systems propagate this estimate from their finite generators to entire root planes. The Steinberg relation

$$[\, I + cE_{01},\, I + E_{13}\,] = I + cE_{03}$$

recovers the missing root direction. Consequently the N-orbit of w has uniformly bounded diameter. Since $w \perp \mathcal{H}^N$, the bounded-orbit projection argument bounds $\|w\|$, and (3) finishes the proof.

More explicitly, if M is the finite control norm used by the two rank-three systems, the formalized estimate is

$$\|b(I + aE_{03})\| \leq D + 12(6\,|X| + 6)(2M + 2D + 1),$$

uniformly in $a \in A$. The numerical constant is not optimized.

3. Coordinate-block factorization

We now prove the main algebraic input. Write

$$e_{ij}(a) = I + aE_{ij} \qquad (i \neq j)$$

for an elementary transvection.

Lemma 3.1 (Five-move pivot)**.** *Let R satisfy (1). If a column of an invertible matrix contains a nonzero entry a, then at most two elementary row operations and at most three elementary column operations produce a unit pivot in any prescribed position.*

Proof. Choose $x, y \in R$ with $xay = 1$. Adding x times the row containing a to the prescribed row creates an entry whose right multiple by y contributes 1. Two further elementary column operations install this contribution as the chosen pivot; one cleanup operation isolates it. Written as block matrices, the calculation is a product of two left and three right transvections. It uses no stable-range or commutativity hypothesis. □

Theorem 3.2. *Under the hypotheses of Theorem A, every $A \in \mathrm{GL}_n(R)$ can be written*

$$A = g_1 \cdots g_m, \qquad m \leq 2n + 6,$$

where each g_i is an elementary transvection or lies in $\mathrm{GL}_{n-1}^{(j)}(R)$.

Proof. Apply Lemma 3.1 to install a unit pivot at (j, j). Clear the remaining entries in the pivot column by at most n elementary row operations and the remaining entries in the pivot row by at most n elementary column operations. The resulting matrix is block diagonal with respect to $Re_j \oplus \bigoplus_{i \neq j} Re_i$.

The scalar pivot is transferred into the complementary block by the elementary diagonal identity available in a 2×2 corner; the resulting complementary block lies in $\mathrm{GL}_{n-1}^{(j)}(R)$. Counting the two initial left operations, three initial right operations, the two clearing families, the coordinate-block factor, and the final elementary diagonal moves gives

$$2 + n + 1 + n + 3 = 2n + 6.$$

Reversing the left operations gives the asserted factorization of the original matrix. □

Remark 3.3. The fixed coordinate j is chosen before A. Allowing the coordinate block to depend on A would give a weaker statement and would not be sufficient for the uniform quasi-cocycle globalization used below.

4. From relative control to global rigidity

Let R be a finite-type $\mathbb{F}_2$-algebra. Choose a finite set X and a surjection

$$\mathbb{F}_2\langle X\rangle \twoheadrightarrow R.$$

Elementary matrices are functorial in the coefficient ring, so the root estimate of Theorem 2.2 descends to R. Coordinate permutations then give the same estimate for every root subgroup in rank four.

Two larger pieces are used in the assembly: a last-column subgroup and a last-row subgroup. The quasi-cocycle is bounded on both. The stabilized rank-three coordinate group normalizes these pieces. Kazhdan control for the rank-three group converts boundedness on the normalized root system into boundedness on the coordinate group, provided the ambient representation has no invariant vectors. This is the only place where the quotient by the trivial representation enters.

Proposition 4.1 (Rank-four assembly)**.** *Under the finite-type and characteristic-two hypotheses above, quasi-cocycles are bounded on each fixed rank-three coordinate block of* $\mathrm{E}_4(R)$*, for unitary representations without nonzero invariant vectors.*

Proof. The six root estimates obtained from Theorem 2.2 bound the quasi-cocycle on the last-row and last-column groups. Let H be the complementary rank-three coordinate group. For $h \in H$, normalization of the two root groups and (2) show that $b(h)$ is uniformly almost invariant under a finite Kazhdan set. Property (T) for H forces $b(h)$ to be uniformly bounded. This is the standard relative-to-global step in the Shalom–Mimura strategy; the formalization supplies all constants internally. □

The passage from a bounded generating subset to the whole group uses the following elementary estimate.

Lemma 4.2 (Products of controlled pieces)**.** *Let b be a quasi-cocycle of defect at most D. If $g = g_1 \cdots g_m$, then*

$$\|b(g)\| \le \sum_{i=1}^{m} \|b(g_i)\| + (m-1)D.$$

Proof. Induct on m, applying (2) at the last multiplication and using that $\pi(g_1 \cdots g_{m-1})$ is an isometry. □

Proof of Theorem B. The finite-type quotient supplies the root estimates, and Proposition 4.1 bounds the quasi-cocycle on the rank-three coordinate blocks in rank four.

The binary Leavitt family gives explicit mutually inverse matrix self-similarities. These transport the rank-four statement to the coordinate blocks required in every rank $n \geq 3$. The elementary diagonal hypothesis identifies the general linear and elementary pieces which occur during transport.

Finally apply Theorem A. Every group element is a product of at most $2n + 6$ factors drawn from sets on which the quasi-cocycle has a uniform bound. Lemma 4.2 gives a uniform bound on the entire group. This is property $(TT)/T$. □

5. The binary Leavitt specialization

Let

$$L = L_{\mathbb{F}_2}(1, 2) = \mathbb{F}_2\langle x_0, x_1, y_0, y_1\rangle/(y_i x_j - \delta_{ij},\ x_0 y_0 + x_1 y_1 - 1).$$

The defining generators themselves form a binary Leavitt family, and the presentation makes L a finite-type $\mathbb{F}_2$-algebra. Pure infiniteness and simplicity give the single-sandwich property [2]. It remains to discuss the unstable diagonal condition.

Proposition 5.1 (Unstable diagonal reduction)**.** *For every $u \in L^\times$,*

$$\operatorname{diag}(u, 1) \in \mathrm{E}_2(L).$$

Consequently $\mathrm{GL}_2(L) = \mathrm{E}_2(L)$.

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5.1. What follows from existing structure.

The conclusion of Proposition 5.1 should not be confused with a new computation of stable K_1. Ara–Brustenga–Cortiñas give, for a regular coefficient ring, the exact sequence determined by $1 - N_E^t$ [1]. For the graph with one vertex and two loops this map is multiplication by -1, hence

$$K_i(L_k(1, 2)) = 0$$

in every degree covered by their theorem, in particular in degree one.

There is also an unstable route already implicit in the structural literature. Ara–Goodearl–Pardo prove that a unital purely infinite simple ring is a GE-ring and that

$$K_1(R) \cong R^\times_{\mathrm{ab}}$$

This is Theorem 2.3 of [2]. Thus $K_1(L) = 0$ makes each unit a product of commutators. The standard 2×2 Whitehead identity places $\operatorname{diag}([a, b], 1)$ in $\mathrm{E}_2(L)$; the GE property then gives $\mathrm{GL}_2(L) = \mathrm{E}_2(L)$. For this reason we claim no priority for the bare equality.

5.2. **The direct formalized route.** The Lean development instead gives a direct tree-and-pencil reduction. Finite prefix codes encode the self-similar pieces of a unit. Elementary row and column moves refine the two codes to a common tree, isolate code-scalar blocks, and reduce the remaining one-sided factor. The output is an actual finite list of elementary matrices whose product is $\mathrm{diag}(u, 1)$.

Proposition 5.2 (Finite factor witness)**.** *For every $u \in L^{\times}$, there exists a finite list $(e_1, \ldots, e_m)$ of elementary 2×2 matrices such that*

$$\mathrm{diag}(u, 1) = e_1 \cdots e_m.$$

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Proposition 5.2 is an existence theorem with a constructive proof object. We do not claim a uniform word-length bound or a complexity estimate: neither is present in the theorem statement.

Proof of Theorem C. The binary Leavitt algebra is a nontrivial finite-type $\mathbb{F}_2$-algebra, its canonical generators form a binary Leavitt family, pure infiniteness supplies (1), and Proposition 5.1 supplies the elementary diagonal condition. Theorem B applies. □

6. Formal correspondence and trust boundary

The public Lean interface is intentionally smaller than the implementation. Table 1 records the exact correspondence between the paper and the checked declarations.

Paper statement	Lean declaration
Theorem 2.2	`PropertyTTPaper.freeCharacteristicTwo_root_hasRelativeTT`
Theorem A	`PropertyTTPaper.coordinateBlock_factorization`
Theorem B	`PropertyTTPaper.finiteType_elementaryGroup_hasTTmodT`
Theorem C	`PropertyTTPaper.binaryLeavitt_elementaryGroup_hasTTmodT`
Proposition 5.1	`KOnePaper.diagUnit_mem_elementary` and `KOnePaper.elementaryGroup_two_eq_top`
Proposition 5.2	`KOnePaper.exists_diagUnit_elementaryFactors`
Corollary 1.1	`PropertyTTPaper.binaryLeavitt_elementaryGroup_hasTTmodT_and_not_isSofic`

Table 1. Paper-to-Lean theorem correspondence.

The synthesis theorem also has the presentation-facing declaration

`PropertyTTPaper.finitePresentation_elementaryGroup_hasTTmodT`, whose hypotheses exhibit a finite free-algebra quotient explicitly. The finite-type wrapper used in Theorem B constructs that presentation internally.

No theorem in this paper assumes soficity, residual finiteness, or any approximation property. The only statement importing the nonsofic development is Corollary 1.1; it is placed in a separate Lean module so that the rigidity dependency graph remains independent. The formalization does not assert historical novelty. Credit claims are prose statements justified by the literature review, not kernel-checkable mathematical propositions.

7. Further directions

The characteristic-two hypothesis enters most visibly in (3), where every root element has order two. Over a field of characteristic p, the corresponding root has order p, and iterating the quasi-cocycle identity gives

$$b(1) \approx b(r) + \pi(r)b(r) + \cdots + \pi(r)^{p-1}b(r).$$

On the root-fixed space this is p times the fixed projection of $b(r)$, with error at most $(p-1)D$. Thus the fixed-space argument has a plausible finite-characteristic analogue. The remaining obstacle is to replace the characteristic-two rank-three control and sign-free coordinate symmetries uniformly. We do not claim such a generalization here.

The algebraic factorization theorem suggests a separate question. For which natural classes of simple or purely infinite rings does the single-sandwich condition hold, and when can the coordinate block in Theorem A itself be controlled by elementary subgroups? A satisfactory answer would turn the present synthesis theorem into a broad rigidity criterion rather than a theorem whose most memorable instance is a Leavitt algebra.

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